

# **Not a voluntary, sustainable, or scalable outcome: Resilience in urban settings of fragility in sub-Saharan Africa**

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**Abstract**

This article focuses on urbanization in settings of fragility, addressing the research question of how new and older settlers manage natural resource use and competition in increasingly urbanizing settings of fragility. Based on two case studies of intermediary sub-Saharan African cities, we find that new and older settlers demonstrate everyday resilience in managing their shared compounded, persistent precarity. This precarity involves constant difficulty in securing water amidst three larger contextual forces: urban poverty, absence of inclusive governance and transparent and fair distribution, and a dynamic landscape of interventions by multiple actors to cope with this precarity. We further find that this resilience was somewhat efficacious in mitigating against the rise of major violence between new and older settlers, though this efficacy may well be highly contingent. We draw on these findings in arguing that resilience may not be a voluntary, sustainable, or scalable solution to increasing urban fragility. Alternative conceptual and practical approaches are finally outlined.

**Keywords**

urbanization, resilience, displacement, sub-Saharan Africa, Praxis

**I. Introduction**

At the end of 2023, the United Nations High Commissioner for Refugees (UNHCR) reported that nearly 110 million individuals were forcibly displaced worldwide (UNHCR, 2023b). This number exceeds the population of Germany and represents a nearly 10% increase in a single year. Internally displaced persons (IDPs) make up the majority of these individuals, with UNHCR reporting that the increase comes as “people [flee] violence, insecurity and the effects of climate change” (UNHCR, 2021). This forced displacement is not only a result of insecurity and environmental change. It can also be a cause for shifts in the security and environment-related community dynamics in the host area. In particular, climate-related influxes of displaced persons have the potential to complicate local natural

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resource distribution and related conflict processes, particularly in already-underserved host areas (USAID, 2014).

These conditions are faced by many urban centers across sub-Saharan Africa, which have become the most common sites of IDP resettlement. Internal displacement is a key accelerant of urbanization on the continent. Of the 6.8 million “new displacements” identified by UNHCR in the first half of 2023, 83% are in sub-Saharan Africa (UNHCR, 2023a). And while popular images of the forcibly displaced tend to show residents in designated “camps,” most IDPs settle in urban areas: rented apartments, unfinished buildings, and informal settlements in the peri-urban outskirts of a city (Earle, 2016). In these increasingly crowded and intensely vulnerable urban areas, it is important to understand the ways in which newcomers, for example, IDPs, together with longer-term residents (re)negotiate access to resources, redistribution dynamics, and potentials for conflict (see, e.g., Rigaud et al., 2018).

Guided by this background, our article focuses on urbanization, operationalized minimally as the increasing settlement of (peri-)urban areas, in settings of fragility, for example, contexts of conflict, forced displacement, state decline, poor infrastructure, and/or resource scarcity. We address the key research question of how new and older settlers manage natural resource use and competition in urbanizing settings of fragility? Based on two case studies of intermediary sub-Saharan African cities, we find that new and older settlers of Bukavu and Maiduguri rely on resilience in managing the precarity of their competitive access to water. This precarity involves constant difficulty in securing water amidst three larger contextual forces: urban poverty, absence of inclusive governance and transparent and fair distribution, and a dynamic landscape of interventions by multiple actors to cope with this precarity. The settlers’ resilience manifested in the data showing that, at different levels and together with other actors, they could maintain some functioning state of largely peaceful access to water despite increasing demand and competition. We further find that resilience as such was somewhat efficacious in mitigating against the rise of major violence between new and older settlers, though this efficacy may well be contingent on current levels of resource stress and responses thereto.

Overall, this paper makes important empirical and practical contributions to research and practice at the intersection of climate change, forced displacement, and conflict studies. To clarify these contributions, we first contextualize our research effort by reviewing literature on the aforementioned intersection. With this literature review, we demonstrate the effect of forced displacement in channeling the influence of climate change on governance, resource management, and conflict in urban settings. We also highlight the gap in understanding how this channeling effect is mediated by the everyday responses of people, newcomers, and longer-term residents alike. Second, we present the theoretical and methodological framework of our study, explaining the assumptions and procedures we followed in addressing the research question. Third, we share the findings of this study, which demonstrate the forms of everyday resilience and the actors and factors that seem to mediate its efficacy in lowering the chance of conflict in the two researched areas. Finally, we discuss the implications of these findings for conceptualizing the emergence of resilience as a response, rather than solution, to the tension between forced displacement and resource management and for interpreting the contingent efficacy of this response in mitigating against conflict.

## 2. Literature review

In this section, we review the literature on the intersection between forced displacement, natural resource use competition, and conflict, focusing on (peri-)urban settings and especially in the context of cities in sub-Saharan Africa. We discuss how forced displacement and urbanization can behave as *channels* through which climate can prompt further conflict, in particular regard to natural resource competition and governance in urban areas.

### 2.1. *The climate–conflict nexus: Displacement and urbanization as channels*

Literature has recently begun to emerge about the impact of climate change on conflict, sometimes referenced as the environment–conflict nexus (Martin, 2005). Scholars have identified an impact on previously known drivers of civil war and armed conflict, including weak per capita economic indicators and state institutions (Bergholt & Lujala, 2012). Other studies have indicated that communal conflict is more likely a result of environmental degradation than of large-scale violence, but that these conflicts are likely to take the form of long-lasting, low-intensity conflicts over scarce resources (Buhaug, 2015).<sup>1</sup> Amidst this emerging literature, climate change is increasingly recognized as a factor in migration and conflict (Podesta, 2019). This relationship—between climate change, migration, and conflict—is complex and is likely to be related to multiple factors, such as personal decisions and individual household strategies as well as institutional and systemic socioeconomic contexts (Cattaneo et al., 2019). Increasing migration has been highlighted as one of the pathways that can lead to conflict, though this is not necessarily the case in all political, social, or economic contexts (Mobjörk et al., 2016).

In the Sahel and African Horn regions, some studies have investigated the role of climate change on pastoralist and herder conflicts in rural areas. However, violence and instability have also been observed from climate change-induced migration in urban areas (Null & Herzer Risi, 2016). While IDPs are often fleeing instances of rural conflict, some of these dynamics may follow these migrants as they settle in urban areas (Moser & Rodgers, 2005). Environmental changes may also exacerbate these influxes of temporary or permanent displacement into urban areas (Adamo, 2010). Particularly in settings of fragility, cities may offer spaces of relative security, but the rapid growth and accelerating urbanization that come with a large influx of new residents can severely strain their limited infrastructures and governance (Pech & Lakes, 2017). As of the early 2010s, more of the world's population lived in urban or peri-urban areas. Major cities and intermediary cities have rapidly expanded, yet infrastructure in both is poorer and state services more limited than in a country's most prominent urban centers. Such cities nevertheless serve as key links in a state's economy, and they often evolve into a hub for many of a state's most vulnerable residents (Githira et al., 2020). As Büscher (2018) notes specifically of the DRC,

[After] 20 years of violent conflict, urban centers in Eastern Congo along the borderlands with Rwanda and Uganda have developed from provincial towns into booming cross-border trading hubs and central nodes in (political) economic networks that connect the mineral-rich Congolese hinterlands to the global markets.

These fast-growing hubs are very often the destination of internally displaced individuals seeking an area to resettle, temporarily or permanently. This urbanization, including that caused by environmental changes as discussed above, is likely to continue in sub-Saharan

Africa. Indeed, based on UN population forecasts, Gizelis et al. (2021) note that “less than one percent of the land mass [in Africa] is defined as ‘urban’. Yet, the population that lives in African cities is expected to increase by more than 150% between 2020 and 2050.” In South Africa, Danielak (2022) notes the instructive case of Setswetla, an informal settlement in Johannesburg that is populated with rural migrants and immigrants from across Africa. When the flash floods of 2016 occurred, many settlers there lost all their possessions. Despite this experience of disaster and of subsequent government action preventing reconstruction in the vicinity, Setswetla became larger than ever, not smaller.

Indeed, this case demonstrates the possibilities of conflict arising from the association of climate change, for example, natural disasters like flooding, and urbanization in settings of fragility, that is, increasing settlement to urban settings whose resources and (infra)structural conditions may exacerbate ethnic tensions, xenophobia, and other forces conducive to conflict (Danielak, 2022). Gizelis et al.’s (2021) larger-scale research assessing the association in Africa further confirms the plausibility of this chain of potential effects: Climate change contributes to displacement, which in turn leads to increased urbanization of settings of fragility, ultimately raising the chance of conflict over natural resource use and competition (also see Elfversson & Höglund, 2023; Elfversson et al., 2023). It is not the purpose of this article to critique the theoretical account of this chain of potential effects. Instead, we only accept it as a plausible theoretical assumption to focus on the potential conflict arising from natural resource use and competition in increasingly urbanizing settings of fragility.

Existing research helps further illuminate these claims about the interrelationship between forced displacement, urbanization, and conflict in settings of fragility. Focusing on Central and Eastern Africa, Büscher (2018) contends that forced displacement and urban population increase can alter the political and economic networks of urban neighborhoods in ways that reflect national and regional conflict dynamics or the compositions of social groups, ethnicities, or language use. This alteration likely exacerbates challenges to natural resource distribution and availability and therefore poses risks of igniting intergroup and political conflict (Yanda & Bronkhorst, 2011). Elfversson et al. (2023) further theorize that urban growth, because of altering group relations, has potential to generate *and* be generated through conflict. Gizelis et al.’s (2021) study offers empirical evidence supportive of this theorization, where they found evidence that population growth in peri-urban areas was associated with increased unrest.

This association between urbanization and conflict is the result of both particular conditions and larger structural factors (Moser, 2004). Newcomers relegated to marginal areas and low-income neighborhoods face social exclusion and a lack of basic citizenship rights. They may also be a convenient scapegoat for persistent social issues and poverty (Stark et al., 2019), not least when communal tensions along lines of ethnicity are stoked (Danielak, 2022). Governments sometimes clash with newcomers to urban centers over illegal land occupations, when the latter attempt to assert a claim to rights (Lombard & Rakodi, 2016). Especially in contexts where governments are weak or corrupt, this may lead to political mobilization or violence based on a lack of perceived state legitimacy (Earle, 2016). Particularly where property rights are weak or poorly enforced, socioeconomic competition and perceived differences in social group competition can lead to conflict (Reuveny, 2007). Indeed, even if newcomers do not actually threaten the livelihoods of older residents, a sense of uncertainty or pernicious narratives about them may provoke conflict (Null & Herzer Risi, 2016). This conflict could take a variety of forms: criminal violence among gang

networks, intimate or domestic partner violence, or more organized political group-based violence (Moser & McIlwaine, 2014).

Having outlined this nexus of forced displacement and urbanization and conflict, it is now opportune to focus on how its dynamics unfold in settings of fragility, for example, poor service capacity, state ineffectiveness, weak institutions, and/or resource scarcity. We conclude this section by referring to the state of “persistent precarity” and thus potential for conflict that can arise from increasing urbanization.

## 2.2. “Persistent precarity,” involuntary resilience, and potentials for water conflict

Based on the literature reviewed above, we understand that urbanization could create the potential for conflict. For urban residents, the dynamics of rapid population growth and poor urban and city planning can manifest as a state of “persistent precarity,” where the state is simultaneously absent and consistently evoked by a variety of non-state actors that take over control of resource access. While certain non-state actors can manage resources more effectively when state capacity is low, this creates challenges with transparency and monitoring and often falls short of maintaining crucial sustained dialogue with state authorities (Batley & McLoughlin, 2010). Across many cities of sub-Saharan Africa, urban governance is fragmented and hybrid, “encompassing multiple sites of power where legitimacy and authority are being exercised and contested” (Büscher, 2018). Amidst this murky configuration of actors, resource access may not always be guaranteed, and residents are often placed in a situation of “involuntary resilience” where they must navigate complex governance dynamics to access basic resources like water.

Some scholars have distinguished between two types of state capacity: capacity to maintain security and capacity to provide services (Krassner & Risse, 2014); however, on a local level, these two functions of a state may be linked. For example, in urban and peri-urban areas, communities with less access to resources are often left out of governance decisions and emergency planning (Satterthwaite et al., 2007). In many parts of sub-Saharan Africa, rapid urbanization, displacement, and resource scarcity intersect with local politics of patronage, rent-seeking, and state capture by powerful elites. Political capture by powerful elites is even more common at the local level, where political parties use the city government as an instrument in a larger game of patronage, extortion, and selective development (Pieterse, 2014). In informal urban areas, where formal state authority may be weak, governance is exercised by non-state actors such as warlords, economic big men, customary authorities, youth gangs, etc. Fragmented power groups compete over political, economic, spatial, and social legitimacy and control over the city, resulting in a divided urban society where key actors continuously challenge one another’s legitimacy to govern urban space. These dynamics result in unstable forms of urban governance, which can turn violent, and form the basis for different forms of urban civic conflict (Büscher, 2018).

In the absence of state capacity to provide services, various local leaders or informal groups may engage in an “institutionalized co-production” of services, strategically engaging with state and non-state actors to secure more reliable access (Joshi & Moore, 2004). Over time, people often develop a form of mutualistic “community resilience” in the face of these dynamics, which can even be “engineered” as a relatively stable form of governance (De Florio et al., 2014). Donors, humanitarian actors, and scholars may typically code such resilience as a positive mode of managing natural resource use and competition in increasingly urbanizing settings of fragility. However, this approach to resilience can obscure dynamics

of neglect, corruption, or exploitation that are involved in the urbanization-conflict nexus (Gooijer & Thomasson, 2006; Kaika, 2017). This approach also runs the risk of treating this resilience as sustainable and its efficacy as scalable, both for meeting human needs—or fulfilling resource rights—and for preventing the occurrence of urban conflict.

Our findings challenge this approach. The two cases presented in this article show that resilience is not developed by deliberate, coordinated, or voluntary choice, nor is its efficacy safe to assume as sustainable and scalable. Rather, the findings demonstrate that resilience manifests in a set of stopgap practices that residents undertake in the absence not only of top-down, long-term, systemic solutions to urban development but also of a bottom-up strategy of maintaining this resilience until more durable solutions are delivered. We draw on these findings in arguing that resilience as such cannot be taken for granted to be a mechanism of preventing conflict in urbanizing settings of fragility. We extend this argument by proposing an alternative theoretical approach to inform more critical research of and practice around the urbanization–conflict nexus. In the following section, we explain the theoretical assumptions and methodological procedures we followed in arriving at these findings.

### **3. Methodological approach**

The research reported in this article draws from a larger study, UNITE, which was conducted in 2021 by the International Rescue Committee (IRC) and funded by the United States Institute of Peace. The larger study focused on linkages between land access, water access and conflict, and its results have been reported in an IRC report (International Rescue Committee, 2022). Recognizing the necessity to better understand conflict around natural resource scarcity in urbanizing settings of fragility, this article focuses specifically on responses to water scarcity in rapidly growing informal urban and peri-urban areas of two intermediary cities in sub-Saharan Africa.

The research reported in this article used an embedded “two-case” exploratory case study design (Yin, 2003) in order to select cities that matched the criteria discussed below, which defined the context of the research problem. Each city was treated as a case study and a unit of analysis, in keeping with the assumption that areas hosting displaced people offer good cases to study issues of the urbanization-conflict nexus (see Martin, 2005). Within each city, two neighborhoods most affected by displacement and rapid growth were identified and treated as embedded subunits of analysis. One of the neighborhoods was urban (consolidated informal settlement, central location, higher density), and one was peri-urban (semi-consolidated settlement, expanding into rural areas, lower density). This design allowed the study team to (a) investigate city-wide processes and city-level conditions as well as inter-group dynamics, (b) explore responses to resource scarcity at the neighborhood level, and, importantly, and (c) investigate the relationships between responses, or lack thereof, at the city-level and those at the neighborhood level. The subunits offered us an opportunity for extensive analysis, enhancing the insights of the case study. Common findings derived from the two case studies characterized by different contextual factors expand the external generalizability of the findings (Yin, 2003). In terms of the qualitative methods used in this design, they comprised semi-structured interviews with experts and settlers at the neighborhood and city levels, and focus group discussions (FGDs) at the neighborhood level.

Case study cities were selected based on the four following criteria. First, we restricted our research population to cities in a region where urbanization is affected by climate change. While the relationship between climate change and urbanization through rural-urban

migration is complex, cross-country panel data suggest that climatic change, as proxied by rainfall, has impacted urbanization trends in sub-Saharan Africa but not elsewhere in the developing world (Barrios et al., 2006). While the impact of climate change on the growth of the two selected cities cannot be readily determined, the selection of both ensured they are located in regions where climate change has affected patterns of urbanization. By applying this criterion, we ensured our selected cases are fit for highlighting some of the effects of climate change on urbanization, the water scarcity responses shaped by the latter, and the potential for conflict that may be associated with these responses. Second, we chose cities in settings of fragility. Cities experiencing conflict can have particularly fast growth as urban expansion processes are often further accelerated by armed conflict near urban centers. Third, we focused on intermediary cities experiencing rapid population growth, the main proxy of urbanization used in this article. While many capital cities are set to become megacities in the near future and have been the focus of urban development and governance interventions, much of the urban growth in sub-Saharan Africa has been taking place in intermediary cities. Finally, we selected only from cities where the IRC recently implemented a governance project. Conducting research in communities with which the IRC works allowed the research team to leverage deep connections with communities to access the most marginalized and affected populations whose voices are rarely heard. In each city, two neighborhoods, one central and one peripheral, were identified based on the level of municipal service provision (limited), income level (low), presence of both host and displaced communities, and limited resources (access to land and water).

In the selected Nigerian city of Maiduguri, a “neighborhood” is defined as a designated area with clear boundaries and represented by a traditional leader called the “Bulama.” Bulama are part of the traditional authorities in Nigeria’s Borno state, and they are often considered the most influential individuals within communities. The DRC has a decentralized model of governance with several levels of administrative divisions: province, city, commune, and sector/chiefdom. Decentralization provides the legal entities management and financial autonomy in decision-making. In DRC cities, like Bukavu, communes are divided into quartiers, which are further divided into cells, and these into avenues. The quartiers are not decentralized administrative divisions as they do not enjoy decision-making and financial autonomy; however, they are the local governance entities that ensure participatory local governance at the community level. For the purpose of this study, the quartiers served as the sub-unit of analysis (neighborhoods). We profile each of these two selected studies below.

### 3.1. Case backgrounds

**3.1.1. Bukavu city profile.** Bukavu is located on the southwestern shore of Lake Kivu and is the capital of the province of South Kivu. In recent years, large influxes of displaced people arriving in Bukavu fleeing violence and fighting in the rural hinterland has intensified conflicts and challenged claims on urban space. Bukavu currently has over one million inhabitants. Over the last 10 years, many of the peri-urban areas have been deforested and now experience heavy erosion (Jacobs & Kyamusugulwa, 2018). Justice, security, land allocation, water supply, and other services are not easily accessible by residents, and rapid and uncontrolled informal urbanization has created strong competition over land. Governance structures seem to have limited ability to manage this as protracted armed conflict in the region has fostered the emergence of parallel networks of power and regulation and new alliances

between urban elites, which has given rise to “hybrid governance” or a situation where local power and authority are contested by multiple stakeholders. Climate stressors, that is, increased temperatures, more variable rainfall, prolonged dry spells, or increased frequency of extreme weather events, are anticipated to significantly impact agricultural production in rural settings, which could further drive urbanization in the city. The two, urban and peri-urban, neighborhoods studied in Bukavu were, respectively, Panzi and Ciriri.

**3.1.2. Maiduguri city profile.** Maiduguri is the capital and largest city of Borno State. Its main local government area (LGA) is 1 of 16 that constitutes one of Nigeria’s traditional states, the Borno emirate. Formal and traditional governance structures are intertwined (hybrid governance), as in most areas in Nigeria. On the side of formal governance, public sectors are governed at the State level, under the umbrella of State sector ministries and public agencies reporting to the governor. All State ministries are headquartered in the city of Maiduguri. At the community level, each neighborhood is represented by traditional leaders, called Bulamas. They are the primary channel through which communities are connected to the institutional structure of the government.

The city of Maiduguri has grown rapidly beyond the initial borders of its LGA with a 1.5% yearly demographic growth since 2000. As a result of the regional instability caused by Boko Haram, 2.1 million have been displaced since 2009 with hundreds of thousands of residents of Borno state seeking refuge in Maiduguri and in the camps surrounding the suburban areas of the city. Maiduguri has seen its population more than double from 1 to 2 million with the influx of people displaced from other areas of the state. The large demographic growth in the city of Maiduguri has resulted in the expansion of the city beyond the initial urban area into other LGAs (Jere and Konduga LGA). This rapid growth in addition to the influx of IDPs has made it more difficult to plan and scale up provision of basic services across multiple LGA (IMPACT, 2020). In an area already economically deprived, more than 78% of internally displaced people are living among host communities (UNOCHA, 2016). The two, urban and peri-urban, neighborhoods studied in Bukavu were, respectively, Suleimanti and Goni Kachalari.

### 3.2. Data collection

In Bukavu, the UNITE Research Coordinator liaised with the members of the Advisory Board to introduce the research to the communities in the selected neighborhoods. To ensure voices of heterogeneous groups within the selected neighborhoods were included in the research, Civil Society Organizations operating in the selected neighborhoods and representatives of different social groups were consulted to identify research participants. Due to the COVID-19 pandemic, all conversations were conducted by telephone. FGDs were conducted with community leader’s representative of heterogeneous groups, and civil society members. FGD participants were purposely selected based on their familiarity with neighborhood and/or city level natural resource management (NRM) practices and governance-related topics as well as the needs of residents in urban and peri-urban areas. Semi-structured in-depth interviews were conducted with host and displaced residents in selected neighborhoods—men, women, youth, minority and majority clans, and persons with disabilities—to capture a range of lived experiences. Key informant interviews were also conducted with state and non-state actors, experts and NGO workers, who were selected based on recommendations from IRC teams in Bukavu and Maiduguri.

In total, 18 FGDs and 54 interviews were held with 120 participants in Maiduguri, 38% of whom were women and 62% were men. In total, 16 FGD and 73 interviews were held with 137 participants in Bukavu, 42% of whom were women and 58% were men. Data collection was first conducted in Maiduguri, with the majority of the interviews conducted in June 2021 and some stretching into July and August 2021, followed by data collection in Bukavu in September 2021. The findings developed from these participants' data are presented in three themes in the next section.

## 5. Findings

### 5.1. *Everyday struggle*

In the areas studied, we found limited evidence of displacement and resource scarcity directly increasing the risk of violent conflict. Instead, our findings indicated that members of old and new settlers of the two cities shared a sense of general precarity, where limited availability and accessibility of water added on to a pervasive, everyday experience of urban poverty. As in other similarly underserved urban settlements, respondents narrated a story of everyday struggle to make ends meet. For example, one interviewee in Maiduguri shared that the increase in population has made access to water difficult, meaning that, "everyone needs the water and so water vendors charge more when the demand is high." In Bukavu, one man shared how his family members sometimes "don't eat because of the lack of drinking water," while another shared that "in the community some people go five days without washing" because of water scarcity.

While tensions around increased pressures on water did not escalate to intergroup or identity-based conflict in the researched neighborhoods in Bukavu and Maiduguri, interviewees described exploitative, speculative, arbitrary, preferential, and other non-transparent practices in the management of urban resources that occasionally lead to interpersonal violence. In the face of limited government regulation or oversight, residents are subject to compounding vulnerabilities and exclusions. Women and children in underserved urban areas are particularly affected, disproportionately bearing the costs of water scarcity and limited infrastructure. Particularly because women and children are often the members of households sent to fetch water, they are often exposed to insecurity, exploitation, coercion, or sexual violence while at, and on journeys to and from, points of collection. For example, in Maiduguri, some interviewees reported that children are forced to clean the houses or property of private water borehole owners before being allowed to collect water, and in Bukavu, several interviewees described widespread instances of rape against girls and women traveling to fetch water in dark early morning hours. For example, in Maiduguri, one man described how "children are sometimes exposed to maltreatment, such as asking them to sweep a compound or dump refuse for them before they are allowed to access water in some private homes," and that "security guards in such homes flog children to prevent them from accessing the boreholes."

In both cities, a lack of transparency about ownership and availability often lead to sharp price fluctuations, particularly in times of overall water scarcity. This price insecurity places enormous strain on already cash-strapped households; some interviewees described situations during which they were unable to prepare food because they could not afford to purchase water. Residents often travel long distances to reach water access points and may wait in long queues, consuming a large amount of time in their day and straining the capacity to

perform care duties or income-generating activities. In both cities, interviewees reported instances of interpersonal violence breaking out at these collection points—typically over disputes in price, bribes for preferential access, or queue-skipping, and some reported frequent fights among children at these locations. However, this violence tends to be episodic and situational, rather than an indication of sustained patterns of inter-ethnic tensions or the result of resentment against displaced persons. For example, one interviewee in Maiduguri reported that “inadequate water supply leads to friction during attempts to access water,” which “has led to prolonged tension between families,” including instances in which “some people have been insulted and labeled as witches during abusive exchanges at water points.”

The already-limited infrastructure is also strained by the security situation, and water collection points can be the target for theft. In Maiduguri, for example, solar-powered or submersible pumps at water boreholes are targets for vandalism or theft, and because replacement is often a complicated and costly process, this may decrease or eliminate operability of these water collection points for long periods. In both Maiduguri and Bukavu, many sources of water are dirty or poorly treated, and residents in both locations mentioned waterborne illnesses or health issues resulting from prolonged reliance on contaminated water. These conditions result in an experience of everyday struggle and reinforcing cycles of precarity and vulnerability.

## *5.2. Lack of urban planning and representation*

Both migrants and longtime residents are affected by outdated urban planning that is inadequate for managing urban growth. Massive population increases in informally settled urban and peri-urban areas, primarily driven by conflict and patterns of rural–urban migration, far outpaces the limited plans that exist for the growth of these cities. As such, access to official water infrastructure is extremely limited, and residents have few representative channels to influence urban government. Instead, structures of informal, traditional, or neighborhood-based governance are often the only ways that residents can raise issues or request assistance; in both cities, these structures are typically the most trusted and accountable, but due to traditional views on gender or existing interpersonal networks, they may also under-represent certain identity groups or new arrivals.

Especially for women, minorities, and IDPs, this can be the case at the neighborhood level. In our research, some women without male household members willing to serve as interlocutors reported instances of discrimination, particularly when attempting to engage with traditional leaders. Especially in Bukavu, some interviewees suspected differential treatment by neighborhood or traditional leaders on the basis of being new arrivals or along the lines of tribal or ethnic affiliation. When governance structures are designed such that concerns are primarily raised by male heads of households, they can also insufficiently account for the individual concerns of women. Ad hoc resource governance structures, such as NGO-launched water committees in Bukavu, are key mechanisms for women’s participation—women make up a large proportion of membership, are involved in key policy development and finance functions, and are instrumental in setting water access prices. However, these bodies are typically selected from within the communities that use particular water access points, and with governance processes set up at the same time as the water access points, new arrivals and IDPs can be excluded from representation.

Urban residents in these neighborhoods face challenges when engaging with the stakeholders they depend on for accessing resources at multiple levels. For example, at the “street-

level,” water vendors who purchase water at access points and resell them in more convenient locations are common in Maiduguri; these vendors are also known to hike prices and clog queues at water access points, making household collection of water without a mark-up more difficult. However, these vendors are a key part of securing water access in these communities, and residents often negotiate and build relationships with particular vendors. In Bukavu, water access sites typically employ a manager, and these managers have been known to exhibit preferential behavior toward certain community members, hoard water for their own household use during times of scarcity, or solicit bribes for timely access. In Bukavu, some water access is negotiated at the state level, via subscriptions and private “sub-subscription” arrangements for access to the water infrastructure of state-owned utility company Regideso. Some respondents complained about Regideso’s non-transparent and arbitrary pricing, which is often passed through “sub-subscriptions,” sometimes with an additional mark-up.

At each of these levels, residents must navigate a set of interests and dynamics that are only weakly mediated by structures of authority. Often, local or traditional leaders are called in to solve problems, but only in the case of individual disputes, and they lack the political heft to advocate for long-term solutions or improved urban planning at other levels of government. The findings in both Maiduguri and Bukavu are consistent with previous scholarship indicating that, in contexts of state weakness or state disinterest (and specifically in urban contexts with lack of urban planning which limits access to resources), communities self-manage and individuals are pushed to find alternative ways to access water.

### *5.3. Filling the gap?*

Non-state actors, communities, and individuals fill the gaps left by the state in urban natural resource management, including mitigation and resolution of water conflicts. These practices and approaches are complex—some facilitate access to water, some hinder it; some mitigate the risk of urban water conflicts at the neighborhood level, and some exacerbate it. Often, structures that facilitate access and mitigate the risk of conflict do limit or exclude certain residents, and in the lack of an overarching presence of government authority, the impacts of certain arrangements of informal, local, or ad hoc governance can vary greatly based on the particular situation or context.

In the researched urban areas, the risk of conflict was largely mitigated and managed at the community level. Community-based organizations and individuals proactively engage and mechanisms between neighbors are devised to address the needs in their communities. Much like in rural areas, informal governance structures (like water committees), elders, and trusted community and traditional leaders play a key role in managing conflict in urban and peri-urban areas. The effectiveness of these leaders—such as the Bulama in Maiduguri, the avenue and quartier chiefs in Bukavu—is key to mitigating conflicts. These leaders reside in their communities, are responsive and advocate for community needs, and are also embedded in vertical governance structures which enable them to refer conflicts to formal and traditional authorities when needed. While not uniform, in the researched areas, the closeness of the community leaders with the communities they represent is in contrast with the observed gap between the communities and the government officials.

However, as mentioned before, traditional authorities may not be sufficiently responsive to the needs of women and minorities. Neighborhood-level governance structures may also not be representative of all residents, as might be the case for IDPs in the Bukavu

neighborhood of Ciriri. Water committees were reportedly not inclusive and responsive to IDP needs in some areas of Bukavu, but strengthening their skills might help, as some respondents reported it did with traditional leaders for conflict resolution in Maiduguri.

Several mechanisms for resource access between neighbors and at the individual level also exist. For example, in Maiduguri, some individuals have drilled private boreholes and control access to water this way. Practices around these boreholes vary—in some cases, owners exercise arbitrary and exploitative pricing, while in others, respondents report some private borehole owners acting as benefactors of the community and charging minimal prices for their neighbors or community members to access water. In Bukavu, interviewees discussed a sub-subscription mechanism, by which one household successfully obtains a Regideso subscription based on its ability to pay and proximity to existing water pipes. This household then offers “sub-subscriptions,” by which neighbors or community members collect water from their household for a fee. In some cases, this appears to be a cost-sharing measure among neighbors, while in others, respondents reported significant profit-seeking behavior from subscription holders. Respondents generally criticized Regideso for arbitrary price hikes, non-transparent practices for billing or metering, poor water quality, and ineffective service, which then impacted sub-subscribers alongside primary subscribers. In cases where residents are renting from a primary owner with a subscription, some reported being charged for water access on top of the cost billed to the owner by Regideso.

We also learned of several community-level structures and approaches. In Bukavu, water committees have been set up by NGOs building new access points, which respondents reported as important and effective structures for community governance and women’s participation. However, as mentioned previously, issues with representing IDPs and new arrivals, as well as a possible lack of overall capacity, have occasionally limited the impact of these bodies. Also, in Bukavu, water point managers oversee community access. In Maiduguri, community-based organizations are involved directly in the provision of water and in negotiation with water vendors. In both cities, neighborhood-level leaders and governance structures are involved in the provision of water. In Bukavu, avenue and quartier chiefs play this role, while in Maiduguri, traditional leaders (Bulama) are involved in managing water access for members of their communities.

NGOs and private sector actors fill a critical gap in contexts of poor urban management and urban planning that has not kept pace with population pressure. However, limited coordination of non-government actors with city governments, coupled with the lack of enforcement of government regulation and quality control, results in suboptimal and unsustainable outcomes. This can be seen in the deterioration of water quality and increase in price by a private provider with an initially innovative and successful approach in Bukavu and is also exemplified by the challenges with boreholes in Maiduguri, some of which are initiated and resourced by NGOs. While NGO interventions help, alignment and coordination with the government are needed, as structural inequalities cannot be addressed through fragmented, timebound, and uncoordinated interventions.

In some urban contexts, public–private partnerships are providing encouraging results (Tillett et al., 2020). This appears to be the case with a unique partnership between an INGO, a government agency, and Chinese state-owned enterprises to support water administration in Bukavu. The IMAGINE (Integrated Maji Infrastructure and Governance Initiative for Eastern Congo) project is implemented by Mercy Corps in coordination with Regideso, with the aim of improving water infrastructure and improving community accountability for these systems. Panzi is one of the areas served by this project, and

respondents in this research indicated that IMAGINE has had a positive impact on water provision in the community. In this research, they affirmed that standpipes installed through the project have expanded access, improved water quality, reduced waterborne illness, and lowered the travel time to access water for many community members. Additionally, several respondents indicated that the presence of Mercy Corps seems to have helped stabilize Regideso's pricing and improve its service provision, with one noting specifically that "since the presence of Mercy Corps, Regideso has also had a fixed price," which has "has brought positive changes in the community." This project is a prime example of how a variety of actors have moved to "fill the gap" amidst a lack of reliable state water provision.

## 6. Concluding discussion

The findings presented in the previous section demonstrate that new and older settlers of Bukavu and Maiduguri relied on resilience in managing their competitive use of water. This resilience manifested in the form of continued competition for water without this competition taking a violent form or becoming a factor stoking identity-based conflict. The rise of this resilience occurred amidst the absence of macro-level, systemic plans to address the requirements of increasing urbanization in a setting of already fragile infrastructure and scarce resources. In this setting, the resilience arising was demonstrated to result from the behavior of (a) private individuals, for example, those drilling boreholes and selling water; (b) community-level actors, for example, quartier chiefs and Bulama leaders facilitating access to water; and (c) NGO-supported structures, for example, water committees tasked with building new access points. In this section, we offer our discussion of this resilience, focusing on how critical interpretation of it informs our understanding of the process that new and older settlers may follow in managing their natural resource use and competition in increasingly urbanizing settings of fragility (RQ).

The findings from both case studies illustrated that in the face of the problem of water scarcity, within the larger problem set arising from increasing urbanization of Bukavu and Maiduguri, the process unfolding was one of "organized anarchy" (Cohen et al., 1972). It had multiple specific actors of various methods and interests of participating in it. The solutions each group of these different actors formulated had some demi-regularity but lacked consensus and coordination. And, resilience—in the form of continued competition without violent conflict arising—seemed to emerge not as a deliberate or voluntary choice but as a haphazard outcome of this anarchical process. This process of managing natural resource use and competition in Bukavu and Maiduguri means that its outcome of resilience may be durable for a while; however, we find little logical plausibility to the claim that this outcome may be durable because the very process elements necessitating its emergence are unsustainable. Here, it is important to note that we shifted our attention from the empirically manifested outcome to the process giving rise thereto. This alternative approach extends Elfversson et al.'s (2023) demonstration, based on insights from Nairobi, Kampala, and Addis Ababa, of the greater plausibility of process-oriented approaches to research on urban settings of fragility. We illuminate this point further in our continued discussion.

The solutions formulated in Bukavu and Maiduguri lacked inclusivity, consistency, coordination, and strategic coherence. Although new and older settlers of the two cities shared a sense of urban precarity, this unity of everyday experience did not prevent water use practices from being exploitative, arbitrary, and unfair especially to women and children. Nor was the increasing settlement and resource demand in either city accompanied by the emergence of a

state of principled collaboration for equitable water access and use, but by a state of competition, albeit largely peaceful for now. Little confidence may therefore be warranted in the durability of these solutions, where access to water is facilitated, sometimes, for some but not for others, where water use arrangements remain highly dependent on specific situations, where resource governance mechanisms may not be responsive or representative, and where macro-level planning and interventions are absent. The efficacy of these solutions in producing resilience, that is, not leading to violent conflict, is contingent on current levels of, inter alia, (a) resource scarcity and demand, (b) acceptance, if not tolerance, of individualized, communalized, and NGOized practices, and (c) absence of conflict entrepreneurs or major forces stoking existing tensions. Should the level of one of these critical factors drop, it may well be predicted to bring closer the possibility of violent conflict in Bukavu and Maiduguri (though these factors and others have varying degrees of inducing that possibility).

Even if this contingency persists and the organized anarchy thus keeps producing the resilience demonstrated in the data, it is difficult to assume this continuity—of everyday urban precarity—should be acceptable. This approach of theoretical care and methodological critique is not a new proposal. It is coherent with Martin's (2005) demonstration, based on a case study in an Ethiopian refugee camp, that conflicts in settings of fragility need to be addressed primarily through the forces constituting them rather than the responses to them. Our approach also illuminates Elfversson and Höglund's (2023) key argument that "in order to understand cities' resilience to violence, we need to account for both the mechanisms linking urban growth to violence, and the possible conflict resolution and mitigation mechanisms present in cities."

The application of this theoretical care and methodological critique in approaching resilience further reiterates Vale's (2013) assertion that "[r]esilience can only remain useful as a concept and as progressive practice if it is explicitly associated with the need to improve the life prospects of disadvantaged groups." By way of conclusion, we discuss this explicit association in terms of its implications on the research and practice of resilience within the urbanization–conflict nexus. We should acknowledge beforehand the technical orientation of these implications, which reflects, at least in part, our concern as practitioners with approaching research as a practice-informed and -informing activity. We concede that more scholarly implications may be gleaned from our critique here, conceptually of resilience and methodologically of evaluating it as a proxy to conflict mitigation in urbanizing settings of fragility.

### *Implications for Research and Practice*

As may be clear in this report of our research, we had not chosen a theoretical framework before our data collection. That is, we had not theorized how resilience arises or how it may act as a key deterrent to conflict in settings of increasingly urbanizing settings of fragility. Instead, we were guided by broad, relevant theoretical assumptions that existing literature offered on the potential interrelationship dynamics between climate change, urbanization, and conflict (see Section 2). This approach to theoreticality, or atheoreticality, helped us be more open to a broader segment of the research participants' reported experiences of managing natural resource use and competition in Bukavu and Maiduguri. This theoretical position allowed us to focus not only on the empirical manifestations of resilience in such settings. More importantly, our freedom from strong presuppositions about the efficacy of resilience as a conflict-mitigating behavior guided us in careful interpretation of the ways in which

resilience may arise and may indeed be efficacious in lowering chance of conflict in (peri-) urban settings of fragility. Within this theoretical position, we pursued a high degree of interpretive validity, that is, deliberately keeping the findings in substance and articulation as fairly representative of the data collected as we best could, before offering our theoretical interpretation thereof (see Wiltshire & Ronkainen, 2021).

This approach to theoreticality may well be useful for critical research on resilience and other modes of responses to increasing urbanization in settings of fragility. Neither resilience nor any such modes of response, identified based on their empirical manifestation, can be assumed to be self-generating or sustainably preventative of conflict. This is consistent with Meerow et al.'s (2016) influential review-based definition of urban resilience such that it is a systemic quality rather than individual behavior. Instead, after exploring new and older settlers' responses, the dynamics giving rise to them and the contingencies of their efficacy for peaceful collaboration or violent conflict need to be carefully theorized. This approach to theoreticality, where theorization follows rather than precedes data collection, allows for better potential to capture context-specific nuances, which then (should) influence theory choices so that inferential accounts of the urbanization-conflict nexus may remain realistic, or ontologically plausible, at least in the contexts where this nexus is explored.

Based on our research, we further believe that Cohen et al.'s (1972) garbage can model of behavior (GCMB) may be a good addition to the theoretical toolkit that researchers use to explore this nexus. The gist of this model is that it describes processes constituted by a multiplicity of actors whose process participation methods, interests, and inputs can be episodic, random, situational, and even arbitrary rather than systematic or predictable. Garbage Can processes are posited to flow from a state of "organized anarchy," patterned inconsistency, and most often incoherence in approaching the problems, solutions, and outcomes with which such processes are concerned (Cohen et al., 1972). The plausibility of the GCMB is already demonstrated, albeit modestly, in pertinent global research, for example, on urban responses to the COVID-19 pandemic in the United Kingdom (Burke et al., 2022) and on oscillating trajectories of urban policy change in Brazil (Leão Marques, 2021). As a theoretical model, GCMB is underlabored (see Verstegen, 2018); that is, it is not prescriptive or even descriptive, for example, of what resilience is, what may give rise to it (in settings of fragility), or how it may mitigate against conflict. By GCMB, the process of responding to increasing urban fragility is only posited to involve certain problems (e.g., resource scarcity; increasing demand; new-older settler tensions), solutions (e.g., at the individual, institutional, or systemic levels), and outcomes (e.g., equitable resource use). Process alignment, that is, between the problems, solutions, and outcomes, should be researched in order to determine whether and how the process can be claimed as one that is favorable and sustainable.

There is also clear practical utility to this theoretical approach to understanding natural resource use and competition in increasingly urbanizing settings of fragility. The first practical implication of this approach is that it is not (only) resilience that may be targeted for scale-up through project interventions. Rather, it is the factors that contingently make it efficacious in mitigating against conflict, for example, resource scarcity and demand not being so severe as to cause greater tension among older and new settlers of Bukavu and Maiduguri; individualized, communalized, and NGOized practices being sufficient band aid *for now*; and the absence of major events or forces stoking existing tensions. It is these factors that should be targeted in the pursuit of sustainable, scalable mitigation against conflict in increasingly urbanized and fragile settings. Some possible actions in this vein should, first and foremost, include meso- and macro-level, short- and longer-term involvement by municipal authorities

and national governments to regulate *and* enforce regulations on fair resource management and to plan sustainable solutions to systemic issues of resource scarcity, urban fragility, and informal settlement. We acknowledge that governments may not always have the financial or technical capacities, or even political will, to pursue such actions. Where that is the case, cooperation and advocacy by actors locally, regionally, and internationally should be offered to support those necessary government actions. Also, other intervening actors like NGOs, whether local or international ones with local offices, should ensure their actions support the environmental, institutional, social, and/or individual factors sustaining community resilience, for example, investing in local technical capacities for water sustainability and innovation, in fair and inclusive community-level resource governance structures, and in context-driven programming for inter-group dialogue and social cohesion. Above all, NGOs should coordinate their interventions with municipal authorities and national governments and align them with the latter's long-term urban development priorities. For this purpose, platforms should be built to mainstream such coordination and alignment as well as to include fair and inclusive representation of the people concerned with the outcomes thereof.

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### Data Availability

The data leading to the findings reported in this article are not publicly available due to privacy and ethical restrictions. Inquiries to the corresponding author are welcome.

### Note

1. Environmental degradation and competition for scarce resources like water are not all driven by climate change. But climate change can compound existing environmental issues, such as overpopulation and poor agricultural practices.

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